PT-AS002

AZURESKY PTY LTD PROFICIENCY TESTING SCHEME – HARDGROVE GRINDABILITY INDEX

REPORT NO 01

OCTOBER 2025

LABORATORY CODE: 28

Prepared in accordance with ISO/IEC 17043:2010 and ISO 13528:2022

PT Scheme Coordinator: Sifiso B Nyambi



PT Scheme Manager: Nonhlanhla Msibi



LETTER TO PARTICIPANT

Dear **Sibonisiwe Coal Laboratory Services Main Lab**,

Thank you for your participation in the AzureSky Coal Proficiency Testing Scheme – HGI Round 01, Oct-2025.

All calculations and evaluations were conducted in accordance with ISO/IEC 17043 and ISO 13528. Results are confidential and may only be used by the participant laboratory. Any outliers are indicated by $|z| \geq 3$. The overall distribution of results demonstrates satisfactory interlaboratory consistency.

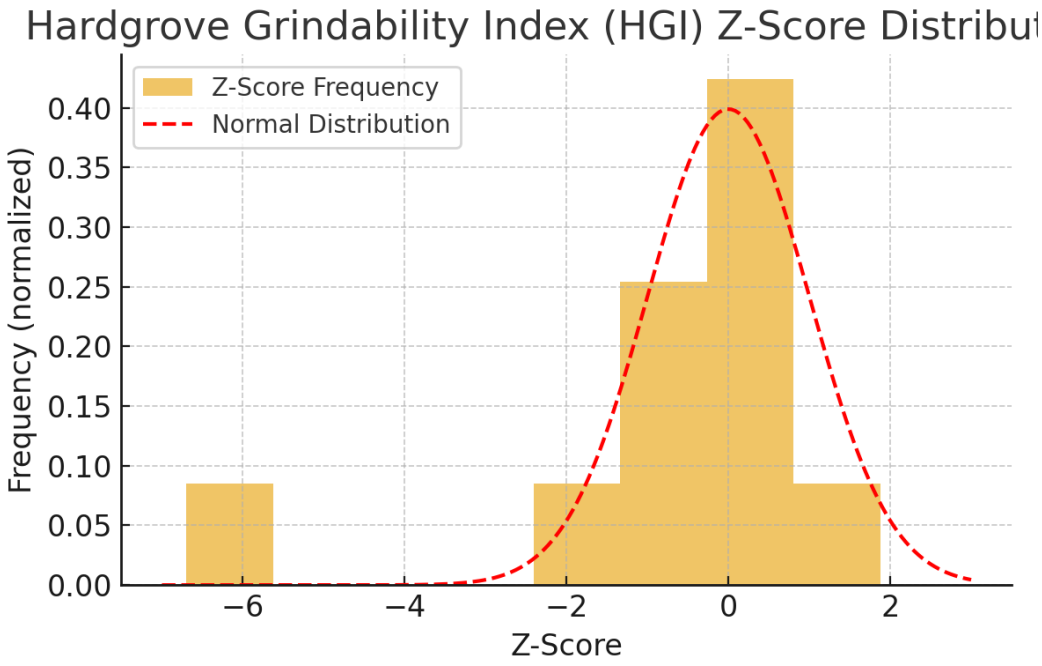
Participants are encouraged to submit any comments or queries regarding the results or operation of the scheme to info@azuresky.it.com or nonhlanhla@azuresky.it.com.

EXECUTIVE SUMMARY

Eleven laboratories participated in the AzureSky Coal PT Scheme for Hardgrove Grindability Index (HGI) during October 2025. All results were treated using robust statistical procedures in accordance with ISO 13528:2022. The assigned value and standard deviation for proficiency assessment were calculated using Algorithm A of ISO 13528:2022.

A total of 11 results were received, with one extreme outlier identified. The overall mean HGI was 53, and the robust average was 52 with a robust standard deviation of 7. The relative standard deviation (%RSD) was 9%. Results followed an approximate Gaussian distribution as shown below.

Interpretation of trends requires a minimum of five data points to ensure a meaningful analysis, while graphical representation necessitates at least two data points to establish a visual comparison.



PT-AS002

TYPE OF SAMPLE USED

The coal used in this proficiency testing round was bituminous coal sampled in Mpumalanga, South Africa

PREPARATION OF SAMPLE

100Kg sample with a nominal top size of 50mm was sourced. It was then stage crushed to -4.75mm using jaw crushers as per the laboratory procedure. The sample was screened using 4.75mm and received on a 3mm screen to further filter the finer material generated during stage crushing. The -4.75mm material was transferred to a homogenizing drum and homogenized for 1 hour. The homogenized sample was transferred to containers with each containing 1Kg of the sample.

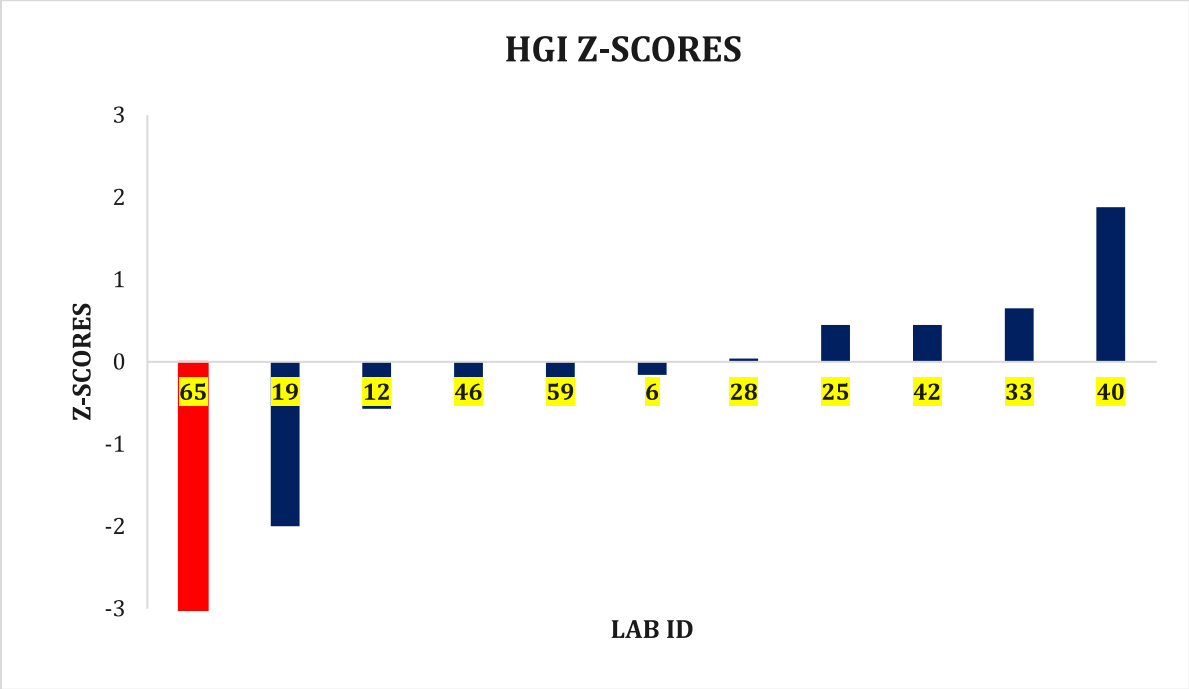
PARTICIPANTS RESULTS

Lab ID	HGI	Z-Score
6	52	-0,16
12	50	-0,57
19	43	-2,00
25	55	0,45
28	53	0,04
33	56	0,65
40	62	1,88
42	55	0,45
46	51	-0,37
59	51	-0,37
65	20	-6,70

Number of scores		11	
Outliers		1	
Mean		53	
Standard Dev		5	
%RSD		9	
Median		53	
Minimum		20	
Maximum		62	
Range		42	
Robust Average		52	
Robust Standard Dev		7	
Uncertainty of Measurement		1	
Upper Reproducibility Limit		66	
Lower Reproducibility Limit		38	

PT-AS002

HGI Z-SCORES



INTERPRETATION OF Z-SCORES

- |z| ≤ 2.0 – Satisfactory performance
- 2.0 < |z| < 3.0 – Questionable result (investigation recommended)
- |z| ≥ 3.0 – Unsatisfactory performance (requires corrective action)

LIST OF PARTICIPANTS

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Eskom Arnot Power Station
Eskom Kendal Power Station
Eskom Matla Power Station
Eskom Research and Testing
Noko Analytical Services Witbank
SABS Secunda
Sibonisiwe Coal Laboratory Services Main Lab
Siza Coal Services Kinross
Siza Middelburg
Umzamo Analytical Services Hendrina
Umzamo Laboratory Services Main lab

CONCLUSION

The range of reported HGI values was 42 units (20 to 62). The robust average and median were consistent, indicating a balanced dataset. One result (HGI = 20) was identified as an outlier with a z-score of -6.7. The general performance of participants is satisfactory, with 91% of laboratories achieving $|z| \leq 2$. The analysis confirms that the PT round met ISO 17043 objectives for statistical validity, assigned value establishment, and reproducibility assessment.